

# Course: Networked Control

## Part I: Course introduction

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# Outline

- Organization of the course
- Goals and topics of the course

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## Organization of the course:

The course consists of about

- **About 50 hours (5 CFU)**
  - About 30/32 hours will cover theoretical subjects.
  - About 6 hours will cover numerical exercises.
  - About 14 hours will be devoted to computer sessions.
- **Theoretical lectures:**
  - Both **slides** and «**board**».
  - There is no «commercial» textbook. **Lecture notes are available.**
- **Numerical exercises** on simple (e.g., scalar) examples.
- **Computer sessions with Matlab.**
  - Analysis and design of networked, decentralized, and distributed control systems.
  - You need your **laptop** with you when required.
  - **Matlab** must be installed.
  - You will also be required to download additional **free software** from the web (e.g., Yalmip).
- Slides, lecture notes, past exams with solutions, and related references will be made available to registered students through WeBeep, <https://webeep.polimi.it> .
- We have 4 hours a week:
  - Tuesdays, 17:15-19:15, room 25.0.1.
  - Thursdays, 14:15-16:15, room IV.

## Organization of the course:

All lectures will be held by

Prof. Marcello Farina

E mail: [marcello.farina@polimi.it](mailto:marcello.farina@polimi.it)

Phone: 02 2399 3599

No specific appointments time. I am very flexible: send me an e-mail to fix an appointment when possible.

Lectures can also be attended by remote on my WEBEX LECTURE ROOM:  
<https://politecnicomilano.webex.com/meet/marcello.farina>

Note that, however, lectures are mostly meant to be delivered in presence.

## Organization of the course:

The **course** consists of three parts:

- I. Introduction and theoretical background
- II. Decentralized and distributed control structures
- III. Networked control systems (NCS)

The **exam** consists of two main parts.

- Regarding the course Part II a **project work** will have to be completed and presented – possibly in groups - in an oral discussion. The project consists of the design – using the tools studied during the course – of distributed and decentralized controllers for a selected benchmark example. During the discussion the knowledge of the students about the topics covered during the course will be also evaluated.
- Regarding the course Part III, a **written test** will be taken, that includes numerical exercises and questions on the covered topics.

# Outline

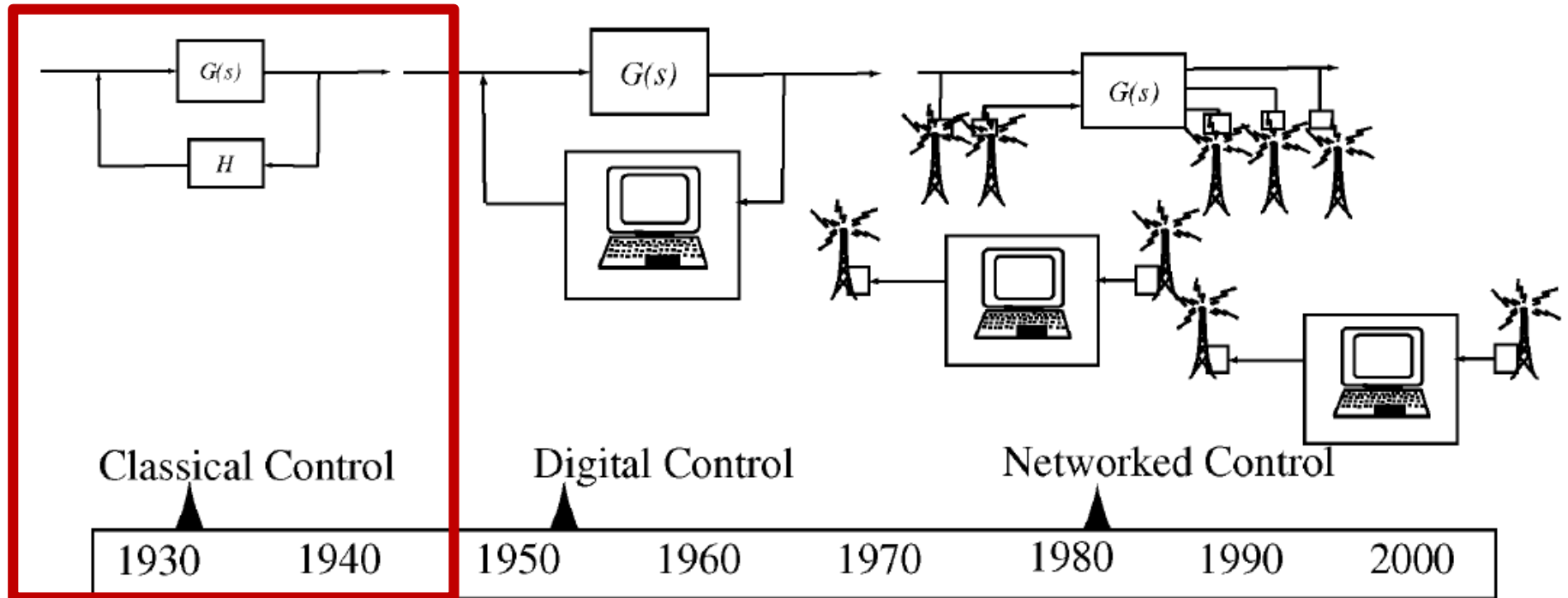
- Organization of the course
- Goals and topics of the course

What is this course about?

Issues arising in the **design and implementation** of  
**control systems**  
characterized by the presence of  
**components arranged in a networked**  
**configuration**



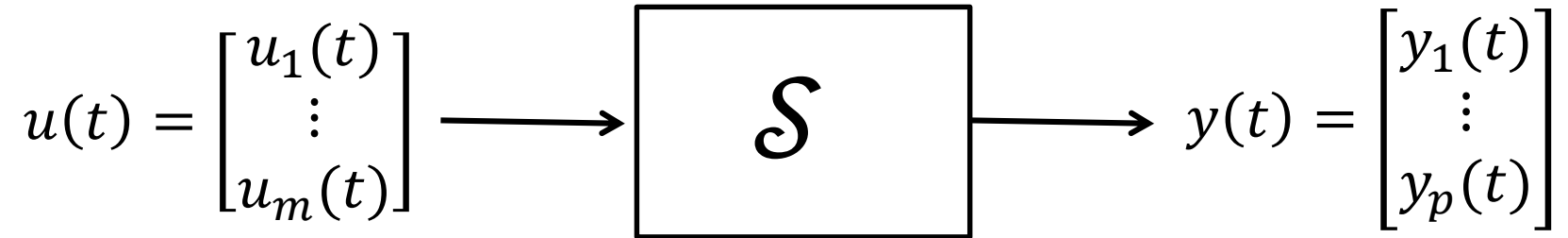
## Let us take an historical perspective



From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

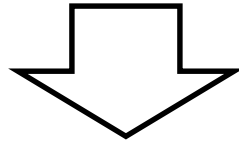
# Classical control problem

Consider an *intrinsically continuous-time* (possibly MIMO) plant  $\mathcal{S}$



Control problem:

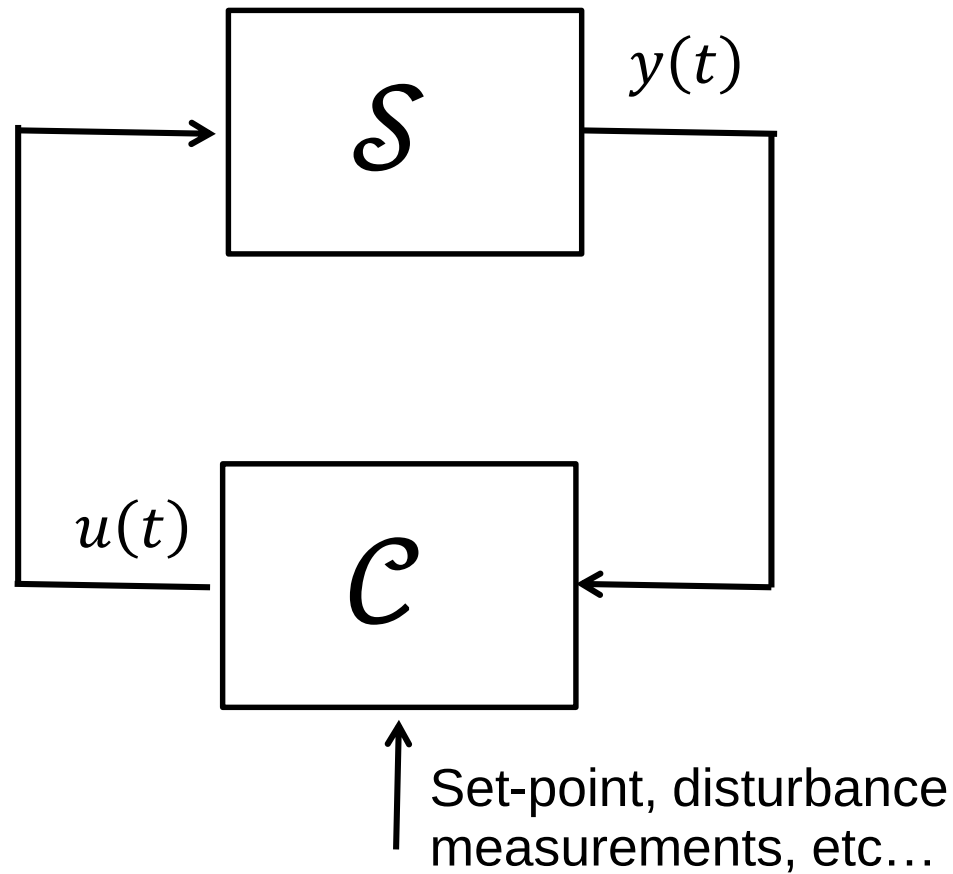
- Steer the output vector  $y(t)$  to a given set-point
- Stabilize the system/equilibrium point
- Optimality requirements (cost function)
- Convey robustness/ disturbance rejection properties
- ...



Design of the controller  $\mathcal{C}$

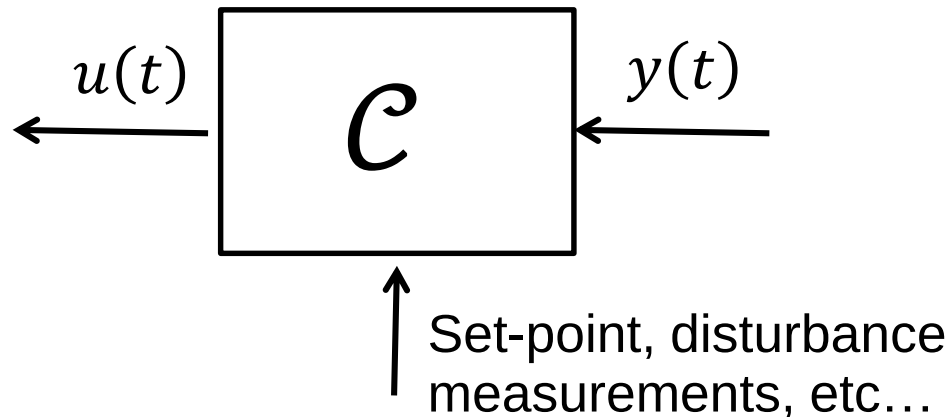
# Classical control problem

Overall feedback control system

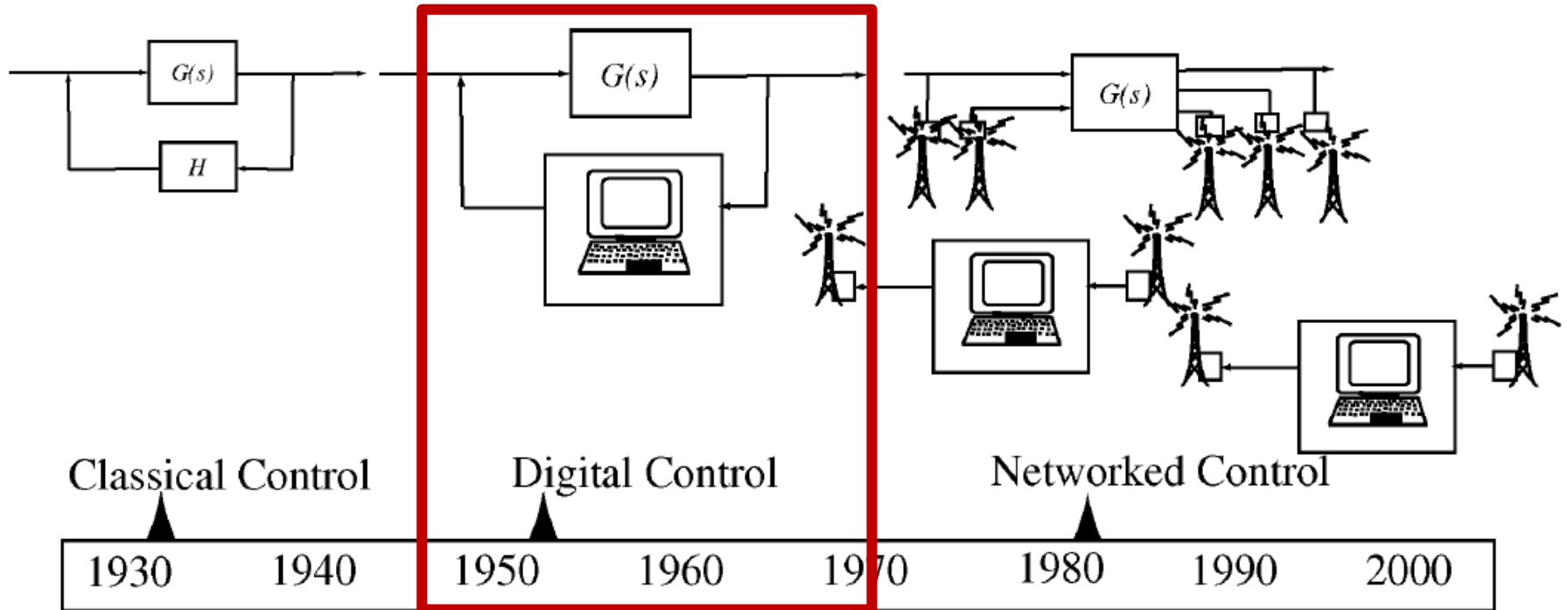


# Classical control problem

- From 1930 to 1950 solid theoretical foundations were established (mostly for SISO systems, frequency domain methods)
  - Controller  $\mathcal{C}$  was traditionally realized using
    - Analogue circuits
    - Mechanical systems
    - Hydraulic systems
- $\Rightarrow$  *continuous-time system*



## Historical perspective

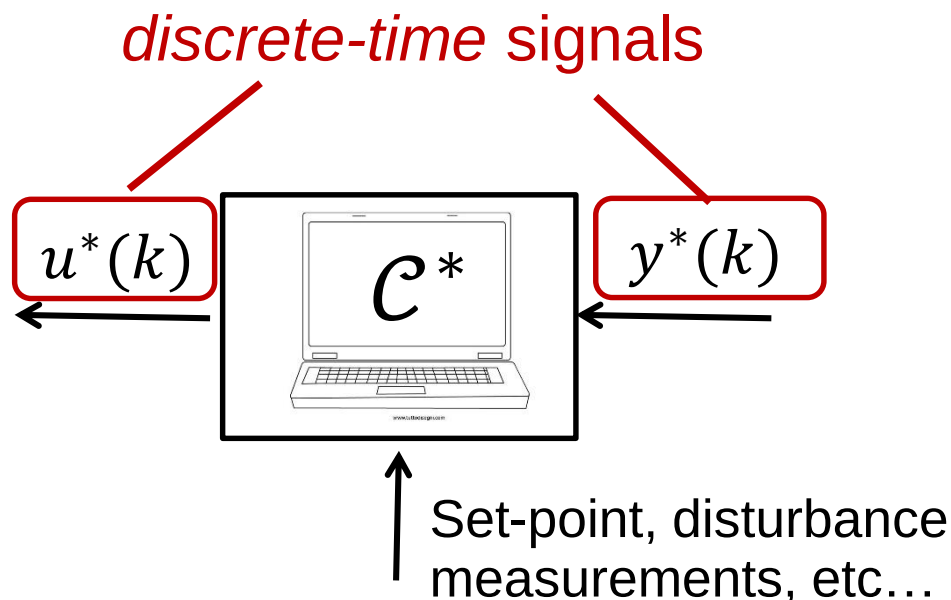


From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

# From classical to digital control

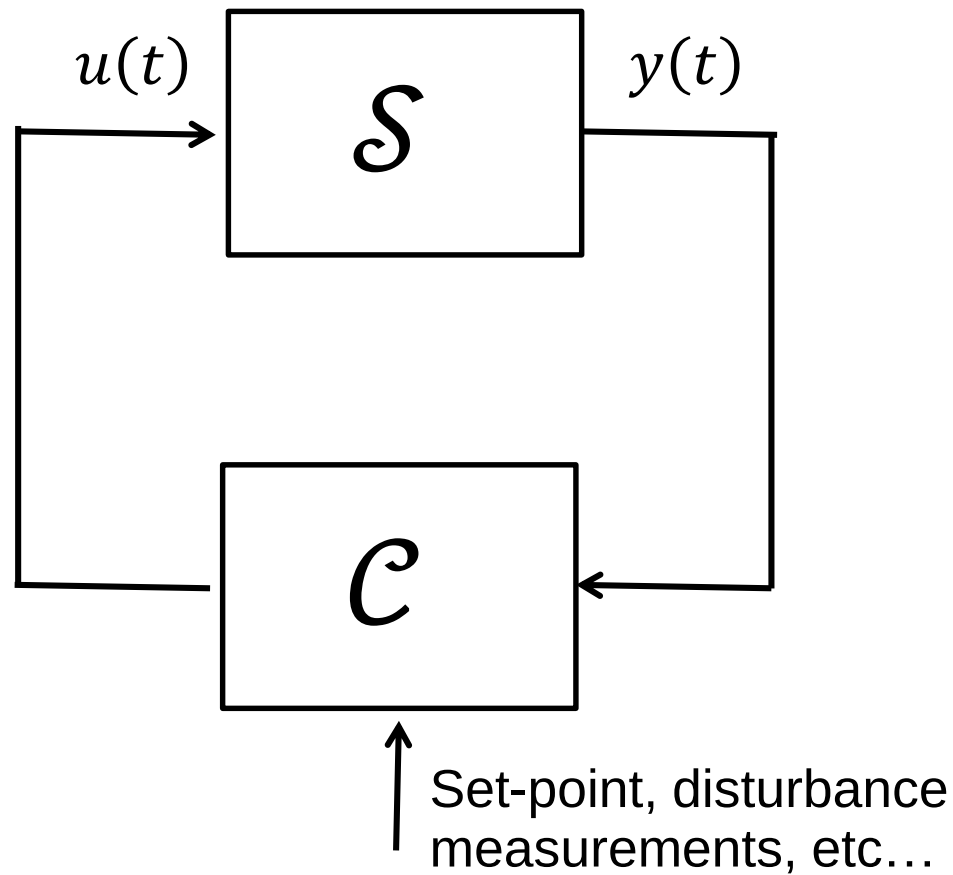
Starting from 1950 there has been a growing interest in the use of digital computers as instrumentation for feedback control

⇒ Controller  $\mathcal{C}^*$ : *discrete-time system*



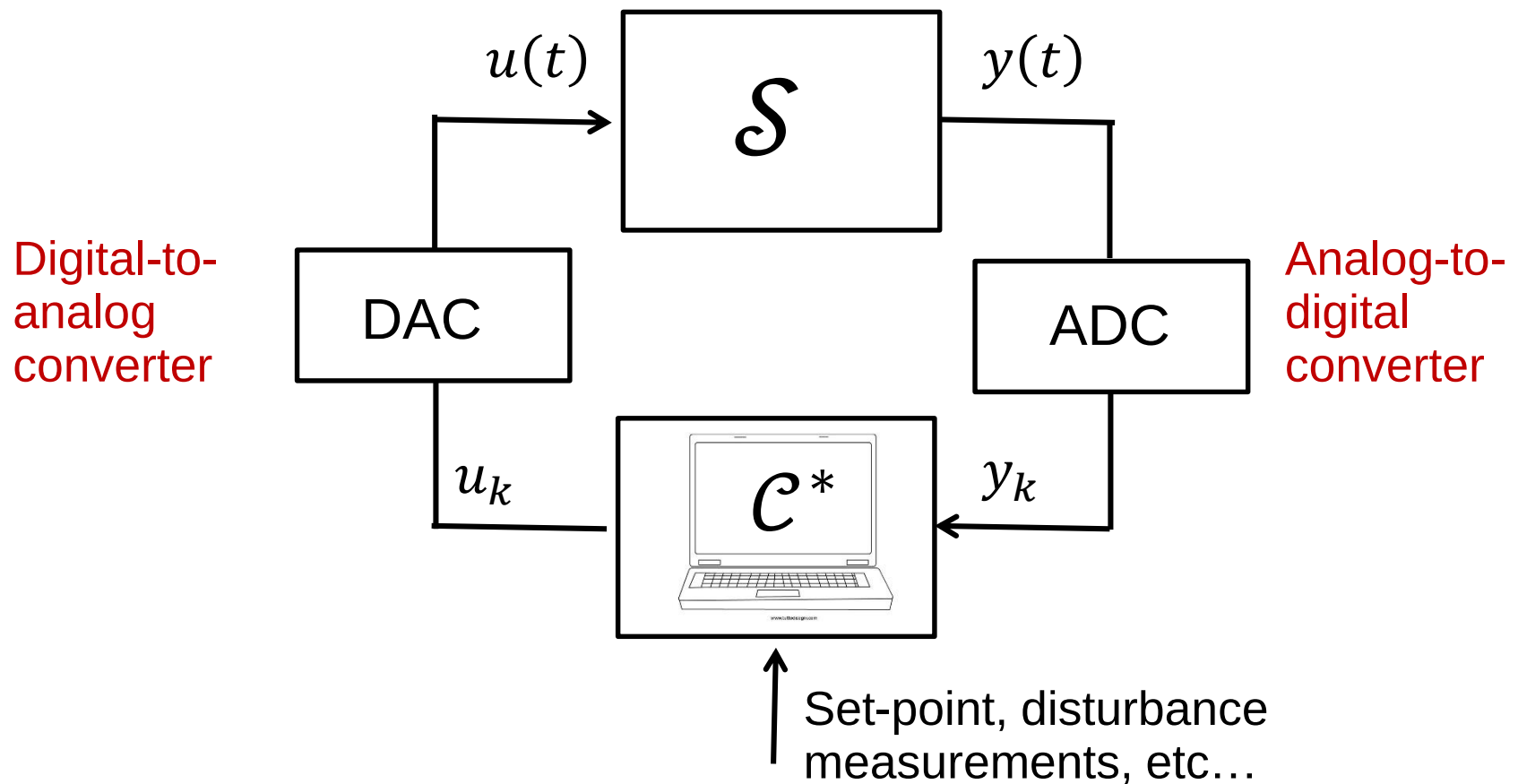
# Digital control

## Overall feedback control system



# Digital control

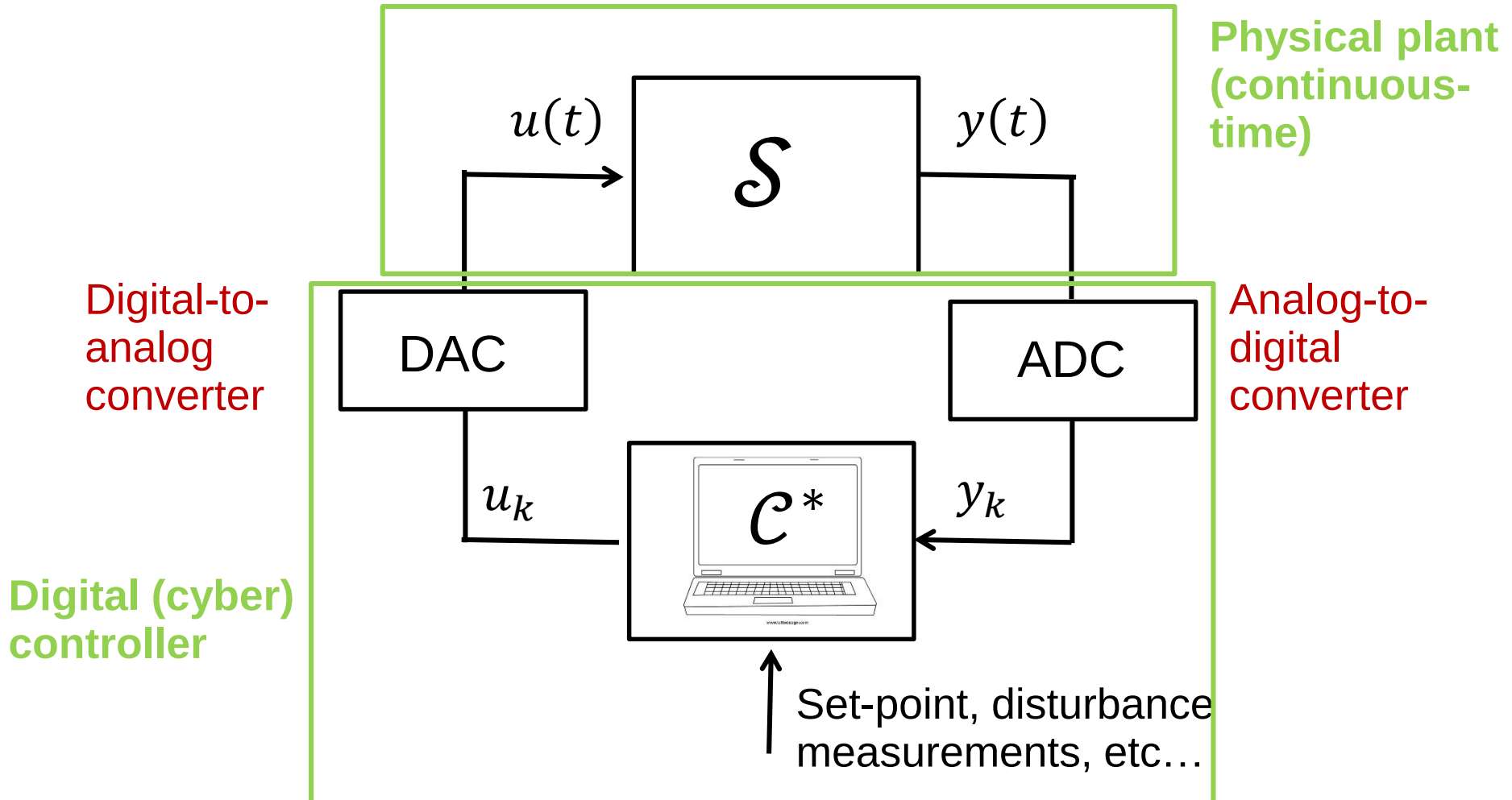
## Overall feedback control system





# Digital control

## Overall feedback control system



# Digital control

Issues:

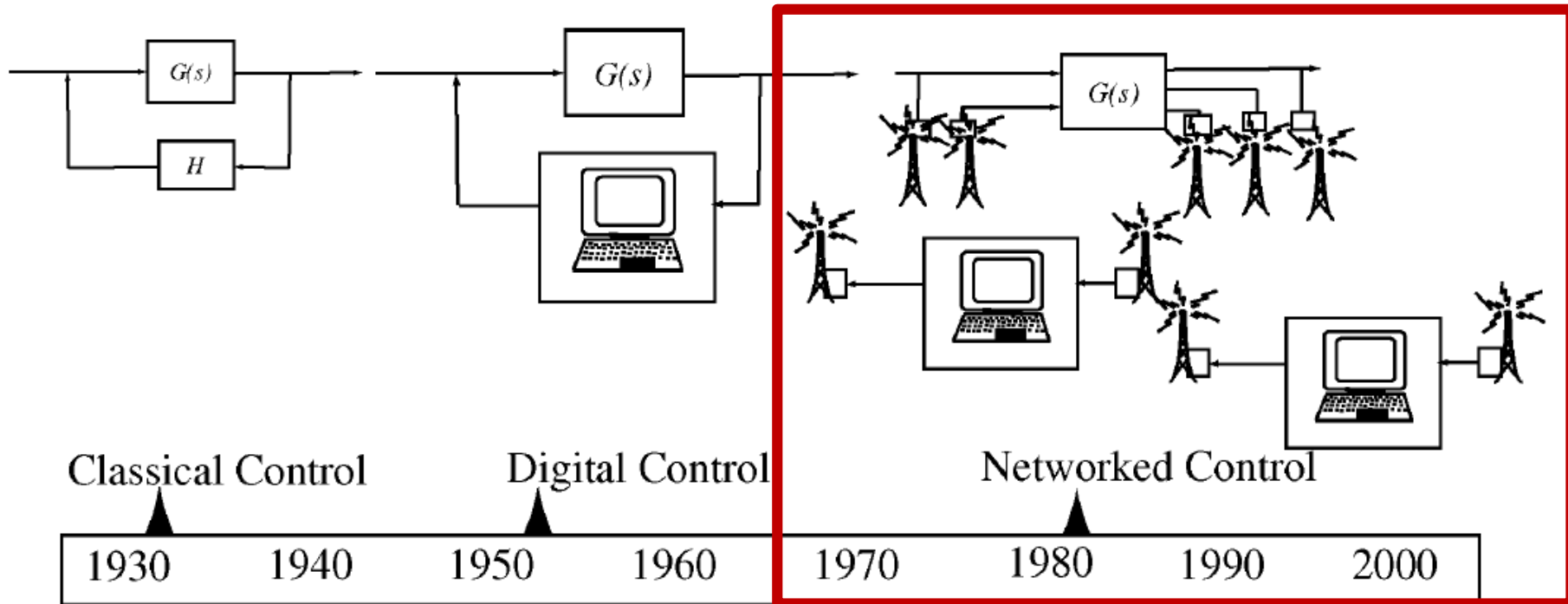
- Effect of discretization/quantization
- Choice of the sampling rate
- Choice of the word length
- Compensation of phase lags

Digital control systems are nowadays also referred to as **cyber-physical** systems.

This terminology is especially used in the computer science area, where emphasis is given on:

- Software/hardware implementation
- Sensors
- Communication issues

## Historical perspective



From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

## From digital to networked control

Up to this point the main underlying assumption for application of (digital, possibly MIMO) control is that

the control system can be

- **designed** in practice **without computational limitations**, i.e., the size (scale) of the system does not prevent the control system to be designed.
- **applied without practical restrictions and without significant performance degradation** with respect to the nominal conditions.

# From digital to networked control

Since the 80's control application have expanded, including plants and systems with **distributed components and/or with large scale**.

**Systems with distributed components:** systems where some parts are not co-located, geographically sparse, or having a modular structure (i.e., they can be regarded as a **network of different subsystems/components**).

## Examples:

- Airplanes/Cars/Motorcycles
- Chemical processes
- Electric networks
- Fleets of agents (autonomous vehicles on ground, air, and sea)
- Irrigation networks
- Manufacturing plants
- ...

# Examples

## Control systems in cars

Present-day cars may be characterized by a huge number of

- **Sensors:**
  - **Pedals** (accelerator, brake)
  - Steering wheel sensor
  - Encoders: four wheels angular velocities
  - Inertial measurement unit (IMU): 3D accelerations + pitch, roll, and yaw rates
  - Pressure in the braking system
  - Suspensions length
  - State of all the actuators (throttle valve, air flowrate in the engine)
  - Lambda sensor: air/fuel ratio
  - Emission measurements
  - Wheel pressure
  - Cabin temperature sensors
  - ...



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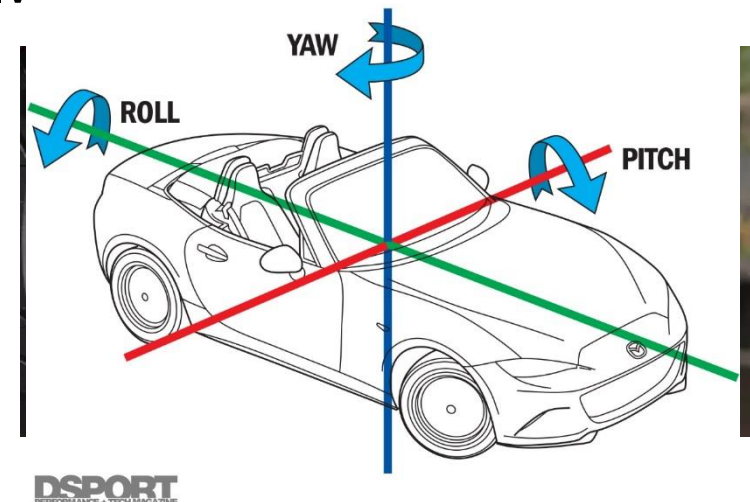
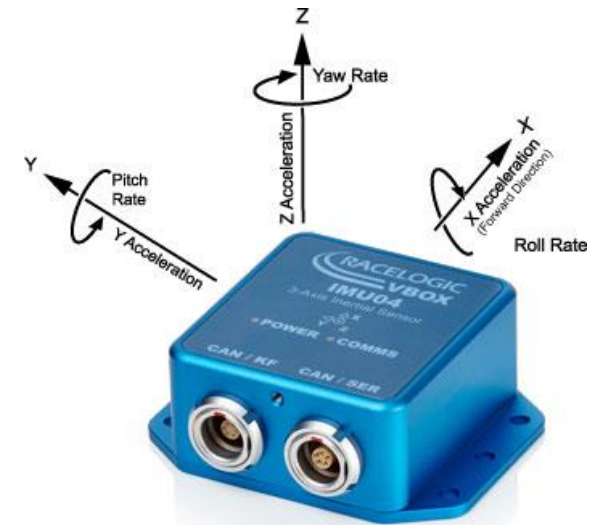
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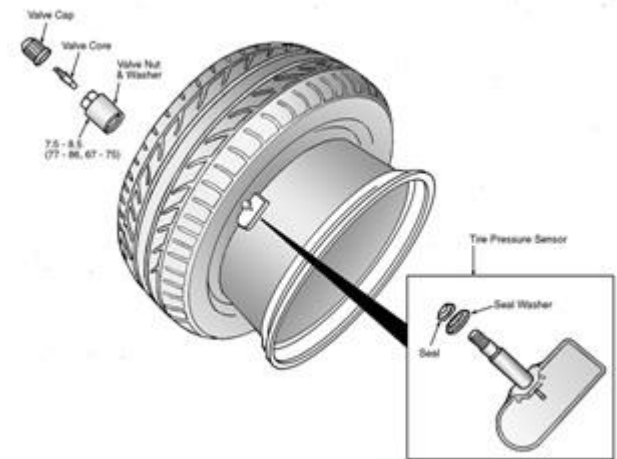


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  - **Wheel pressure**
  - Cabin temperature sensors
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# Examples

## Control systems in cars

Present-day cars are characterized by a huge number of

- **Actuators:**
  - **Throttle valve:** air flow into the engine
  - Fuel injector
  - Hydraulic modulator (braking pressure actuators)
  - Semi-active suspensions
  - Air conditioning actuators
  - Brakes
  - ...



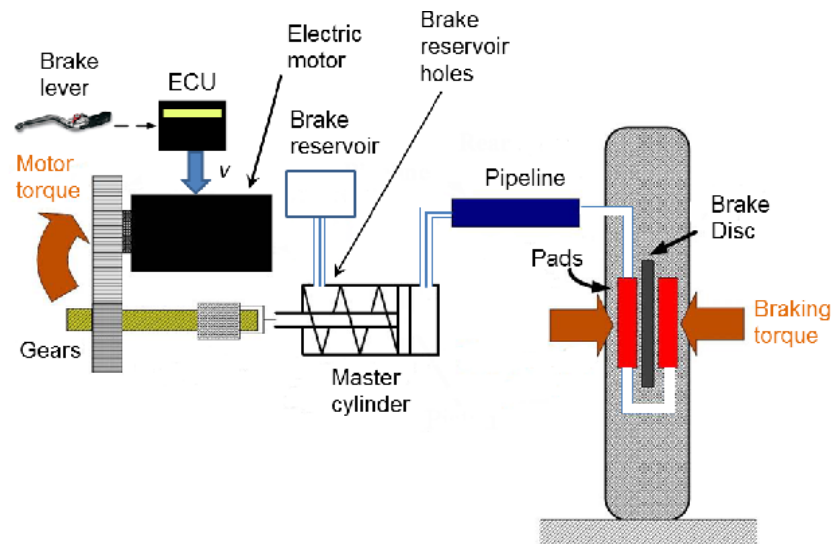
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  - Fuel injector
  - Hydraulic modulator (braking pressure actuators)
  - Semi-active suspensions
  - Air conditioning actuators
  - Brakes
  - ...
- **Control systems:**
  - **ECU** (Engine Control Unit): to follow traction torque setpoints given by the acceleration pedal or other vehicle control systems, while controlling emissions. Talks with actuators: throttle valve, gas recycle, air injection.
  - **ABS** (anti-lock braking system): to follow deceleration setpoints given by the braking pedal without incurring in slips.
    - Reduce stopping distance.
    - Maintain Steerability
    - Maintain lateral stability.

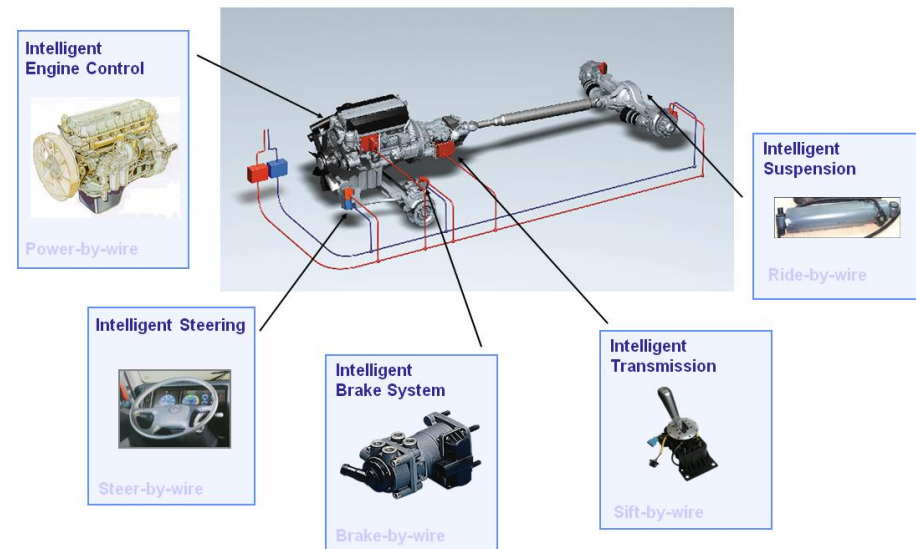


# Examples

## Control systems in cars

Present-day cars are characterized by a huge number of

- **Control systems:**
  - **VCU (Vehicle Control Unit):** High-level dynamic control system that manages active suspensions, ESP (electronic stability program) or ESC (electronic stability control).
    - Cruise control: maintains constant longitudinal speed
    - ESP: Helps improve cornering and control. It can sense differences between the driver's intentions and the vehicle's directions during turns.
    - TCS: Traction Control System. Prevents vehicle's wheels from spinning excessively while on slippery surfaces. Complementary to ABS.
  - **Dashboard display** control system: mainly for processing and displaying data (for the driver) from sensors
  - **Anti-collision assist**
  - **Lane warning**
  - **Air conditioning.**



# Examples

## Control systems in cars

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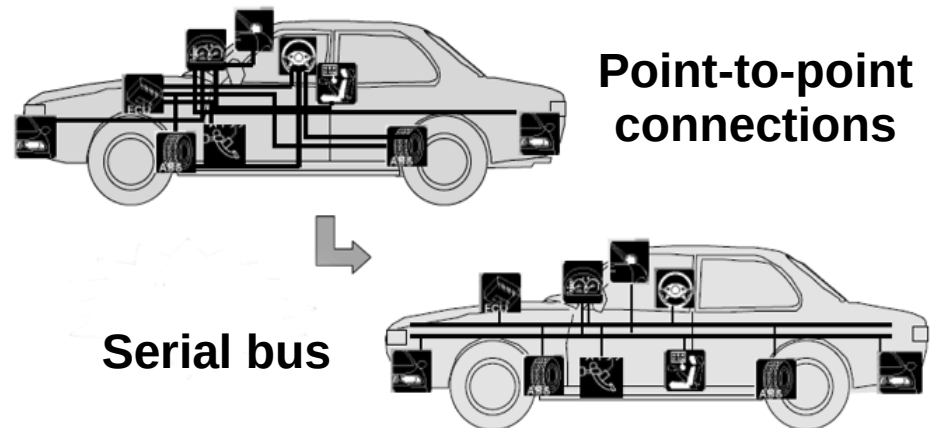
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  - **Air conditioning.**

Why so many different hardware control systems? To reduce complexity, increase modularity/maintenance and since they are generally produced from different companies!

# Examples

## Control systems in cars

- All these elements (sensors, actuators, control units) must **talk** to each other!
- Wired **point-to-point connections** are impractical
  - Costly
  - Space consuming
  - Messy
  - Not reconfigurable (addition/replacement of devices)
- Present solution: **serial bus** CAN (controller area network). Shared digital network.
- Communication problems, due to the shared network:
  - **Delays** (possibly non-bounded and non-constant)
  - **Package dropouts**





## Examples

## Natural gas treatment plant

## Natural gas:

Nitrogen  
carbon dioxide  
methane  
ethane  
propane  
iso-butane  
n-butane

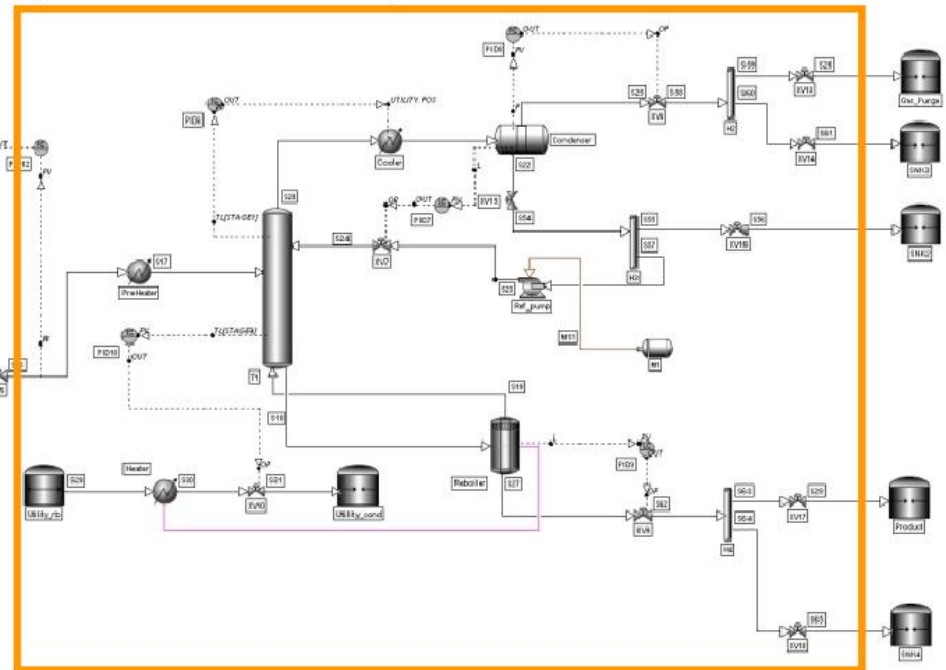
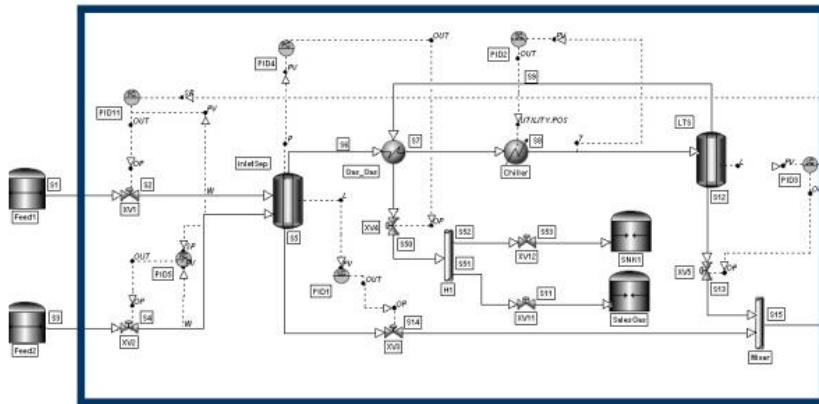
## Refrigeration section

Propane  
iso-butane  
n-butane

Nitrogen  
carbon dioxide  
methane  
ethane

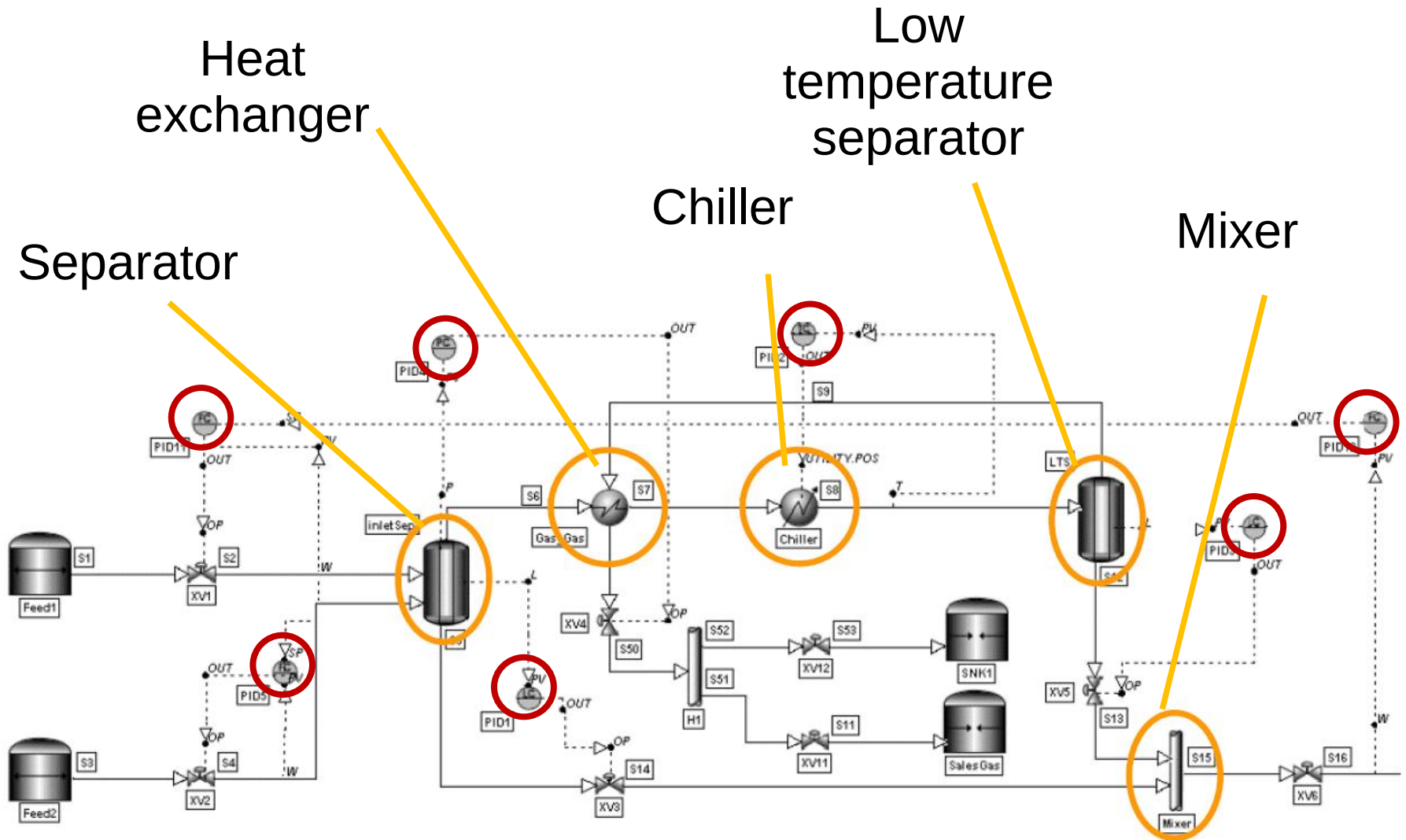
## Separation section

## Pipeline, gas flaring



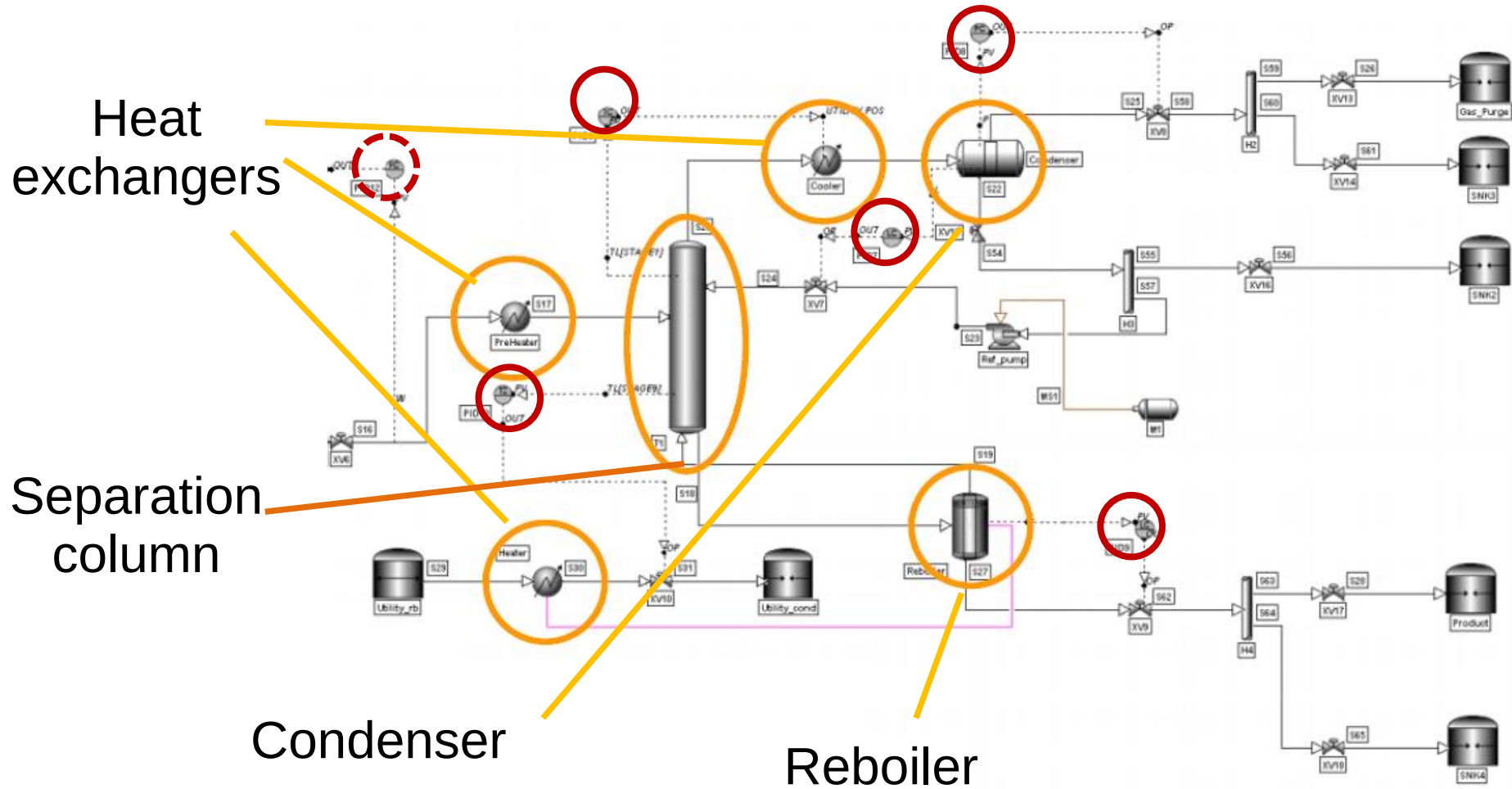
# Examples

## Natural gas treatment plant – refrigeration section



# Examples

## Natural gas treatment plant – separation section



# Examples

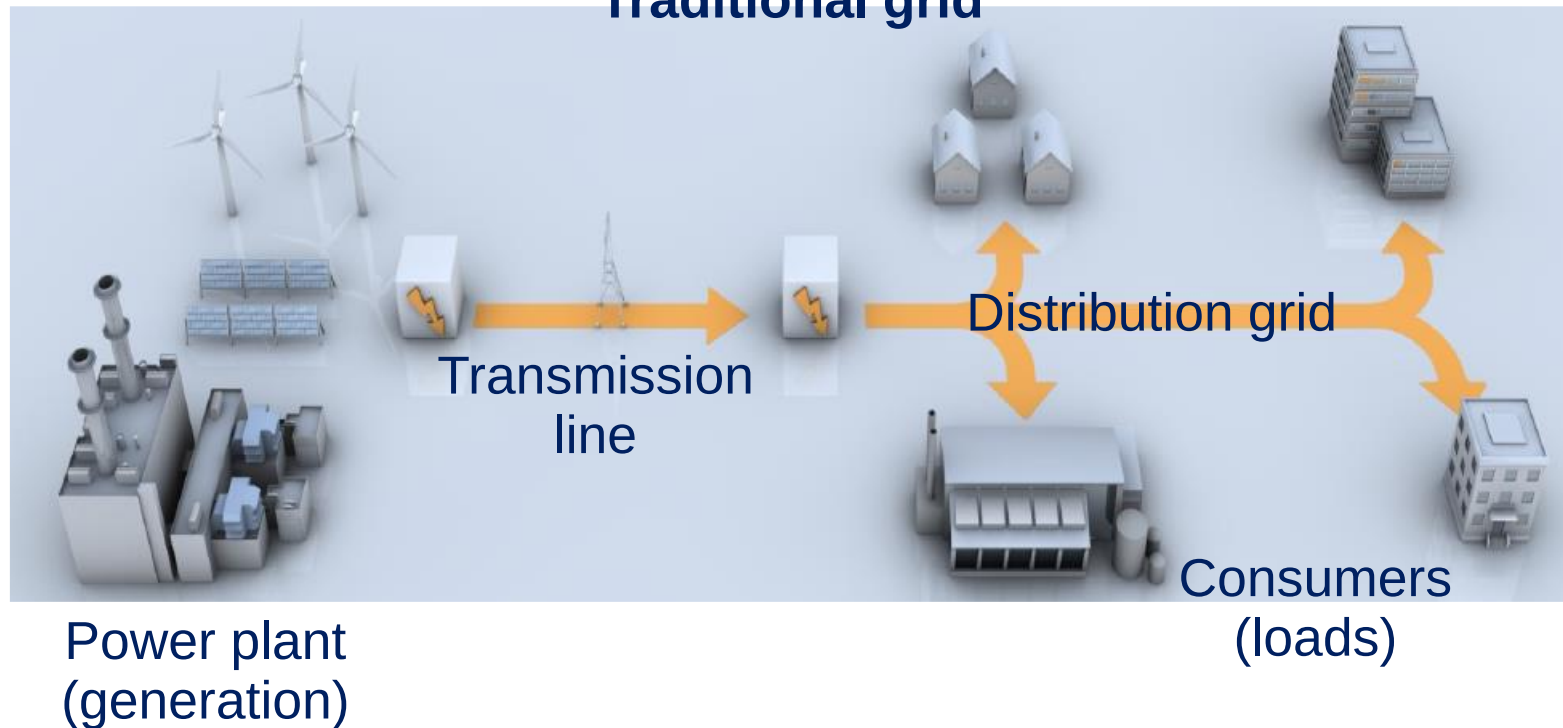
## Natural gas treatment plant

- **Large-scale system**, composed of many interacting subsystems
  - Subsystems are geographically distributed
  - Each subsystem must be subject to control (is itself a complex and critical system)
  - Interactions must be considered for optimal performances
- MIMO system: (**centralized**) multivariable control?
- Challenges:
  - Large control problem dimension – **computational challenge**
  - Can information must be conveyed to the control station efficiently, «rapidly», and «simultaneously» **from the different sensors** and **to the different actuators**? **Information structure constraints** may be present!
- **Present control scheme**: each subsystem is controlled independently (**decentralized** control): loss of performance/risk of instability.



# Examples

## Electric grid Traditional grid



Drawbacks:

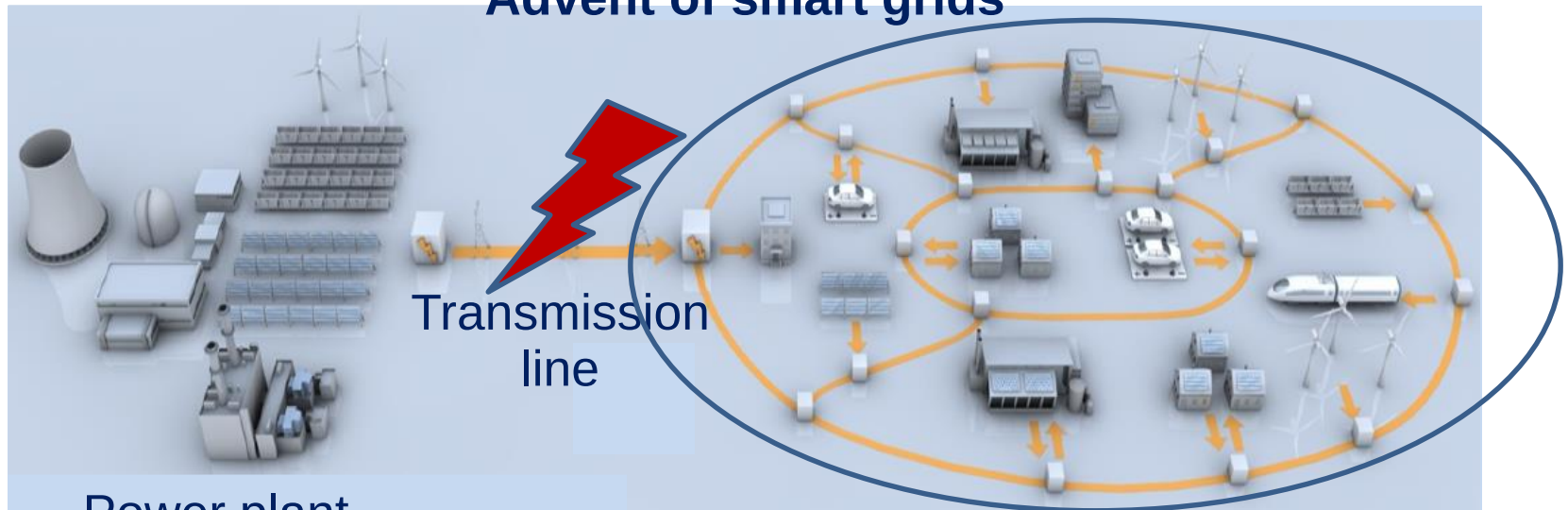
- Few generation plants far from loads
- Large use of fossil fuels

Increasing demand and availability of local (i.e., at consumer level) and «clean» electricity (e.g., PV panels, wind turbines)!

# Examples

## Electric grid

### Advent of smart grids



Power plant  
(main  
generation)

Micro-grid  
(generation+  
loads +  
batteries)

#### Benefits:

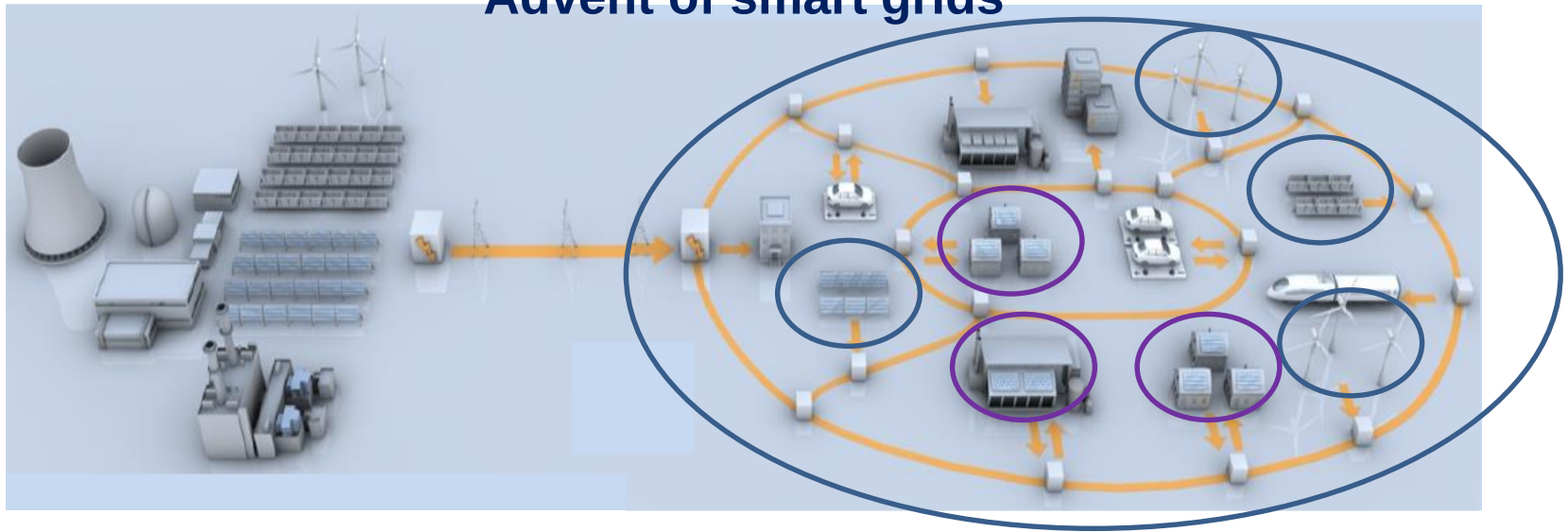
- Distributed generation close to loads
- Increasing use of renewables
- Redundancy
- Modularity
- Possibility of operation in islanded mode (self-sufficient network of distributed generation units)



# Examples

## Electric grid

### Advent of smart grids

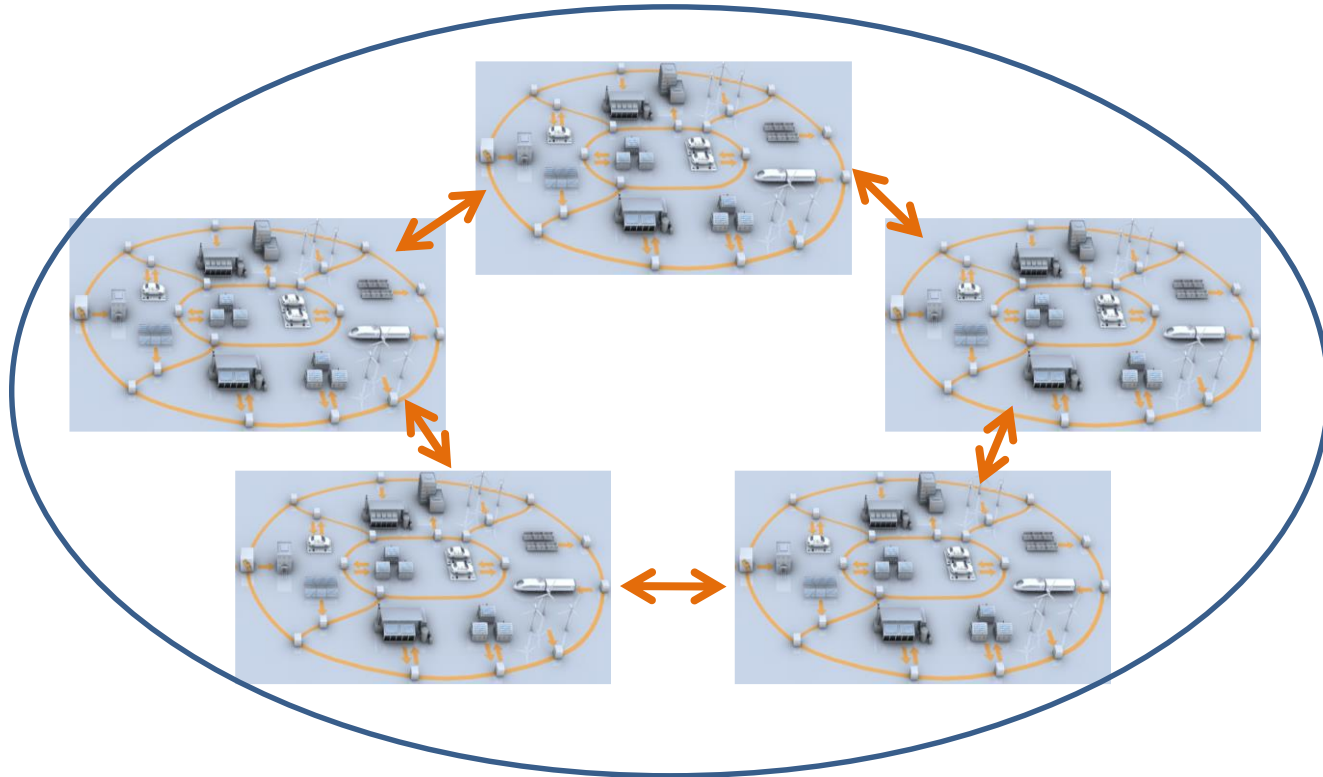


Control and optimization issues at different levels:

- Single **generation unit** level:
  - Stable operation.
  - Reactive power control .
- Single **microgrid** level:
  - Guarantee the fulfillment of energy needs (with possibly controllable loads).
  - Maximize gain from local production.
  - Minimize purchase from the grid.
  - Frequency/voltage regulation (especially in case of islanded operation).

# Examples

## Electric grid



- **Aggregator level:**
  - Cluster of microgrids that take the role of a **(virtual) power plant**
  - Active power reserves
  - Reactive power reserves



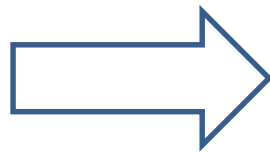
# Examples

## Electric grid

Traditional centralized control structures are not suitable

Need of more complex control architectures

- Control at different levels
- Coordination



Novel **hierarchical** and **distributed** control architectures

Communication

# Examples

## Fleets of underwater autonomous vehicles for sensing

Goal: **design a mobile sensor network** to collect measurements from a field

- e.g., salinity, temperature, chlorophyll fluorescence (i.e., a proxy for the concentration of phytoplankton, bottom of marine food chain, indicator of the state of the marine ecosystem and of the atmosphere)
- optimal measurements: sensors should **move in a rigid formation**.

### Moving sensors are underwater gliders:

- possibly autonomous underwater (e.g., for ocean depths) vehicles (AUV), equipped with sensors for measuring the environment.
- can collect data continuously over periods of weeks or even months.



From: N. E. Leonard *et al.* Collective motion, sensor networks, and ocean sampling. *Proceedings of the IEEE*, 95 (1), pp. 48-74

# Examples

## Fleets of underwater autonomous vehicles for sensing

It is not possible/convenient to consider the **fleet as a single system** to be controlled in a unified (centralized) way:

- **Lack of flexibility** (i.e., with respect with the number of agents).
- **Unreliable** (i.e., with respect to faults at single agent level).
- Possibly very large-scale system (when the number of gliders is large):  
**computational limitations**: control actions must be taken at a centralized level for all the agents.
- Severe **communication limitations**: measurements are taken at a local level, they must be transmitted (instantaneously and synchronously) to the central control station and control actions must be transmitted (instantaneously and synchronously) from the control station to the local actuators.
  - Radio frequency communication (used in other contexts, e.g., land) cannot be used in water.
  - Acoustic communication not practical (distances of kms between gliders and with the central station).

# Examples

## Fleets of underwater autonomous vehicles for sensing

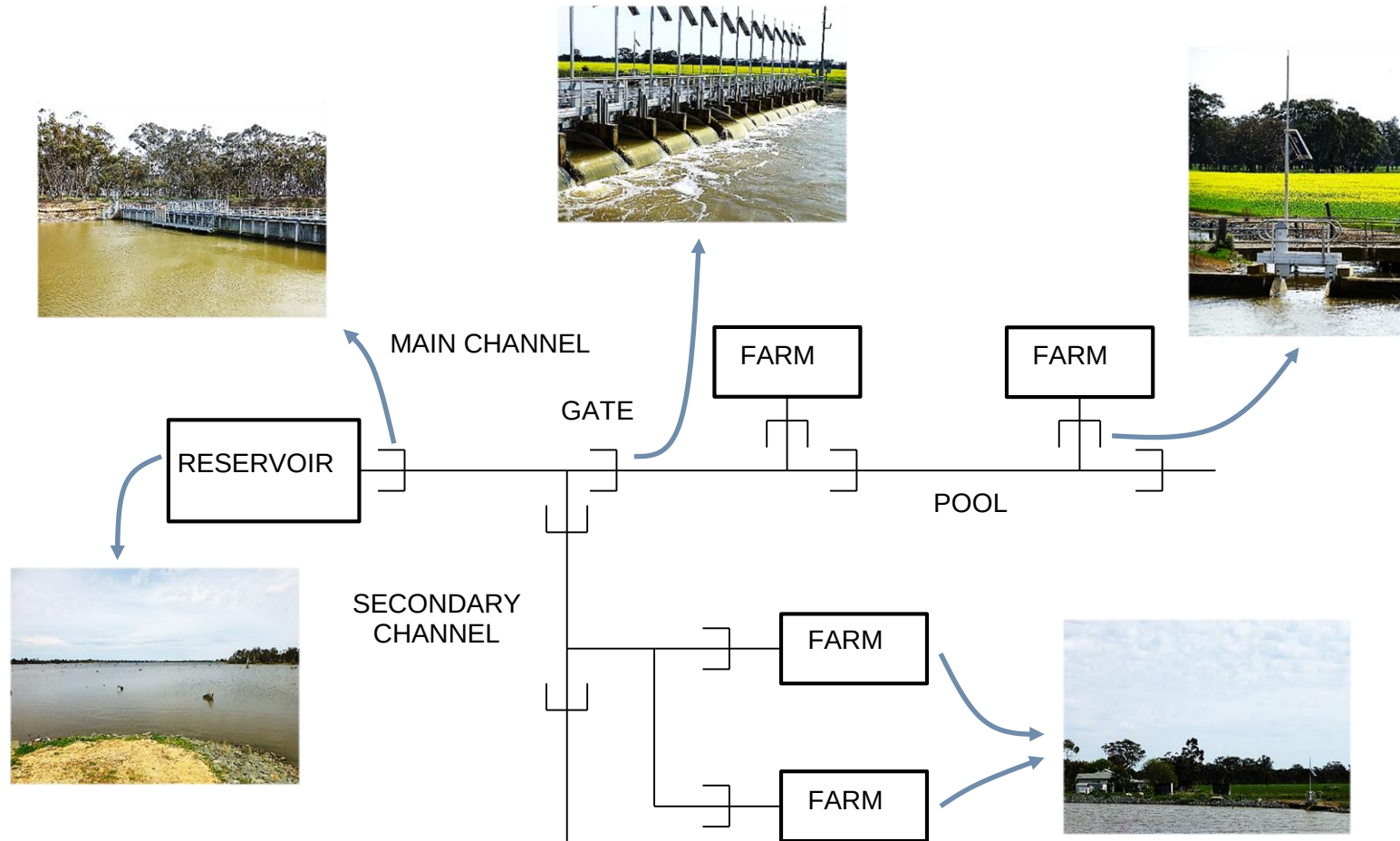
### Adopted solution: 2-level control

- **HIGH LEVEL cooperative control** (implemented in a **single control station**, but possibly also in a **distributed** way):
  - Definition of way-points (i.e., trajectories) using (distributed) artificial potentials, to maintain a formation (virtual body).
  - This allows to **stabilize the desired formation**.
  - **Communication** is available only when gliders surface (periodically and asynchronously), when also
    - GPS measurements are obtained and transmitted to the control station.
    - waypoints are transmitted by the control station to the gliders.
  - **Challenges** (tradeoff):
    - High frequency surfacing for better coordination
    - Low-frequency surfacing for saving energy
- **LOW LEVEL** (local):
  - Heading and pitch control
  - Follow way-points
  - Necessary data collected and estimated locally



# Examples

## Irrigation networks

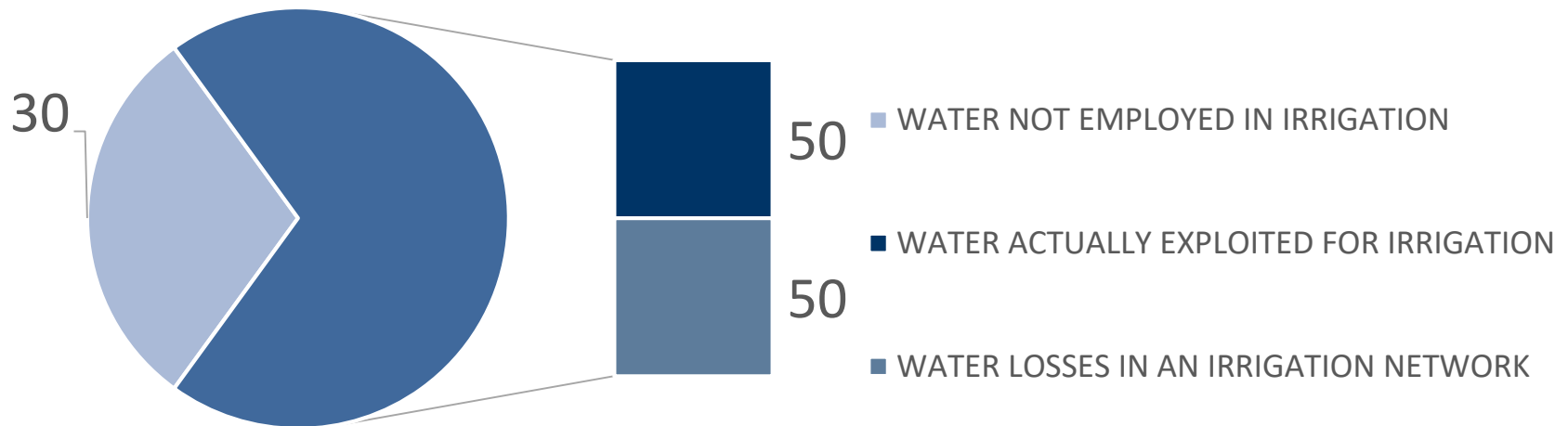


From: M. Cantoni *et al.* Control of large-scale irrigation networks. *Proceedings of the IEEE*, 95 (1), pp. 75-91

# Examples

## Irrigation networks

WATER DIVERTED FROM RIVERS



Large scale distribution losses

- water over-supply
- water under-supply

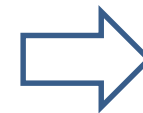
Aquatic ecosystem emergency

Overflowing and water wastage

Distribution losses at farms

- Manual actuation of the gates

Errors and delays in water supply

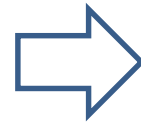


A fully automated control can improve water distribution efficiency up to 85%

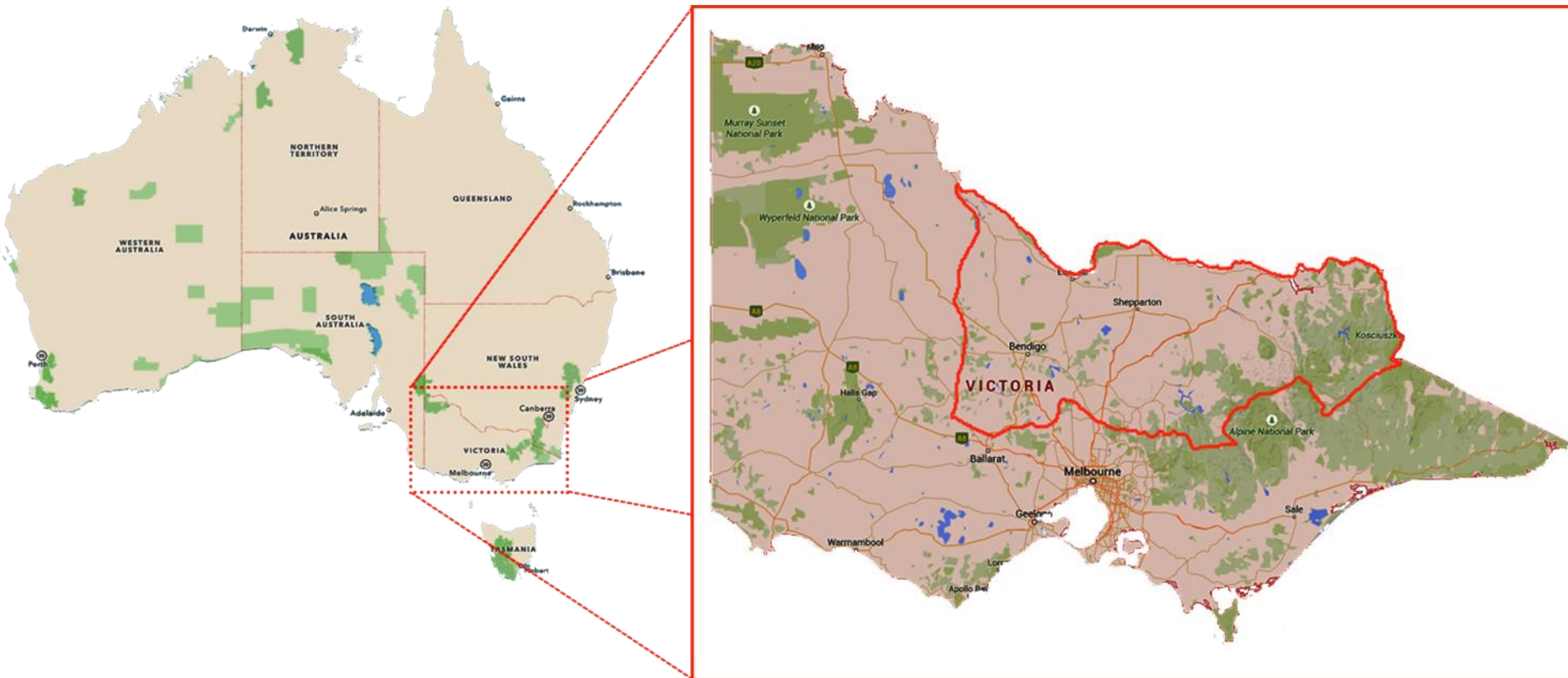
# Examples

## Irrigation networks

- Murray-Goulburn Irrigation district:
  - 7000 Kilometers of channels
  - 68000 km<sup>2</sup> area
  - 21000 farm outlets



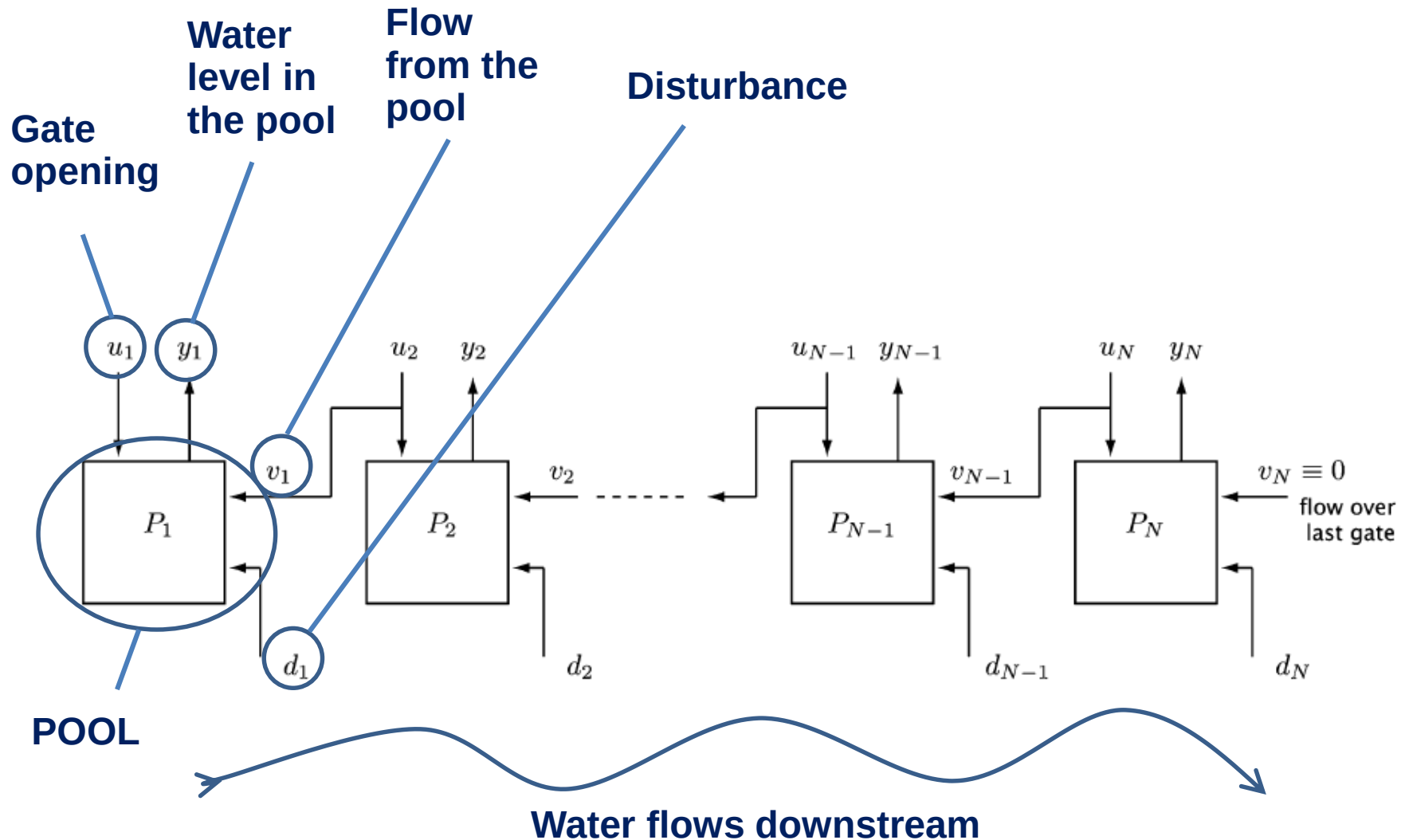
Large-scale system





# Examples

## Irrigation networks





# Examples

## Irrigation networks

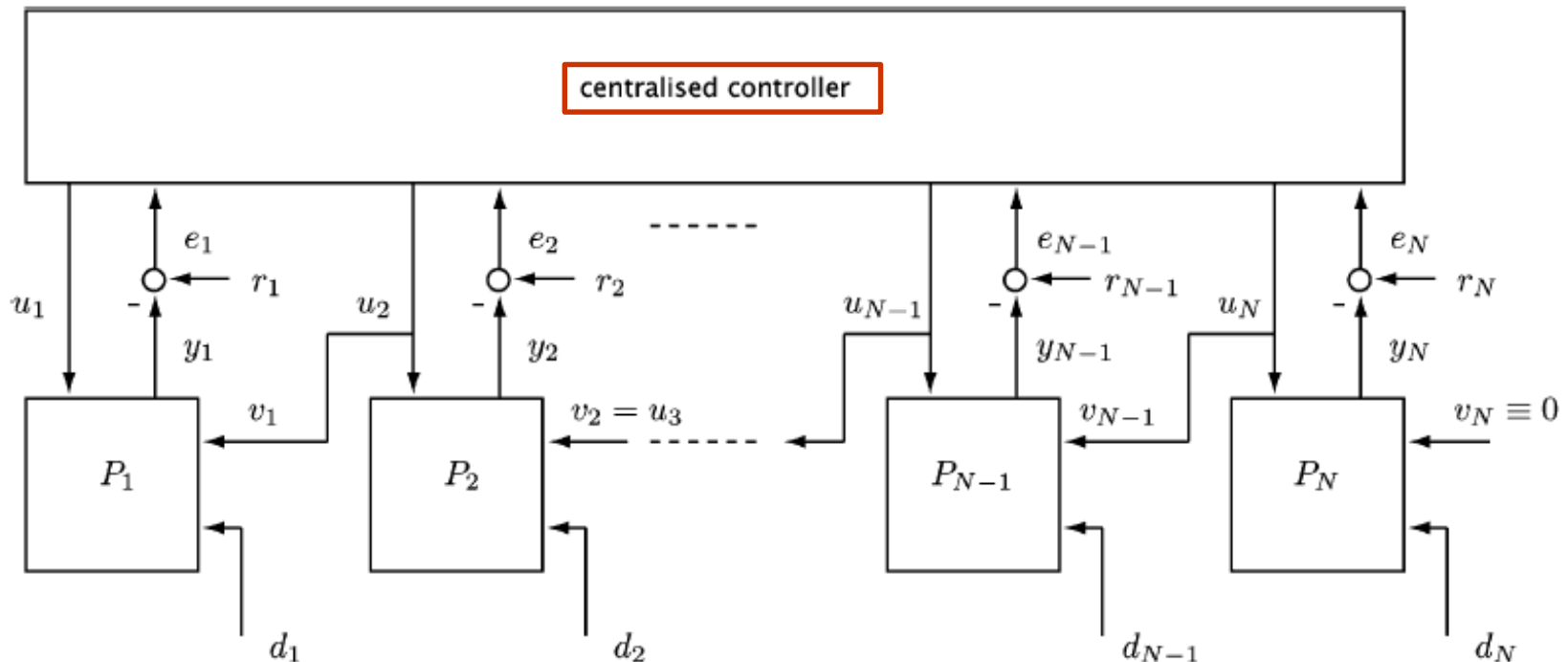
- Centralized computation
- Centralized information structure



Not compatible with a possibly very large-scale system!



Distributed or decentralized solutions are necessary!

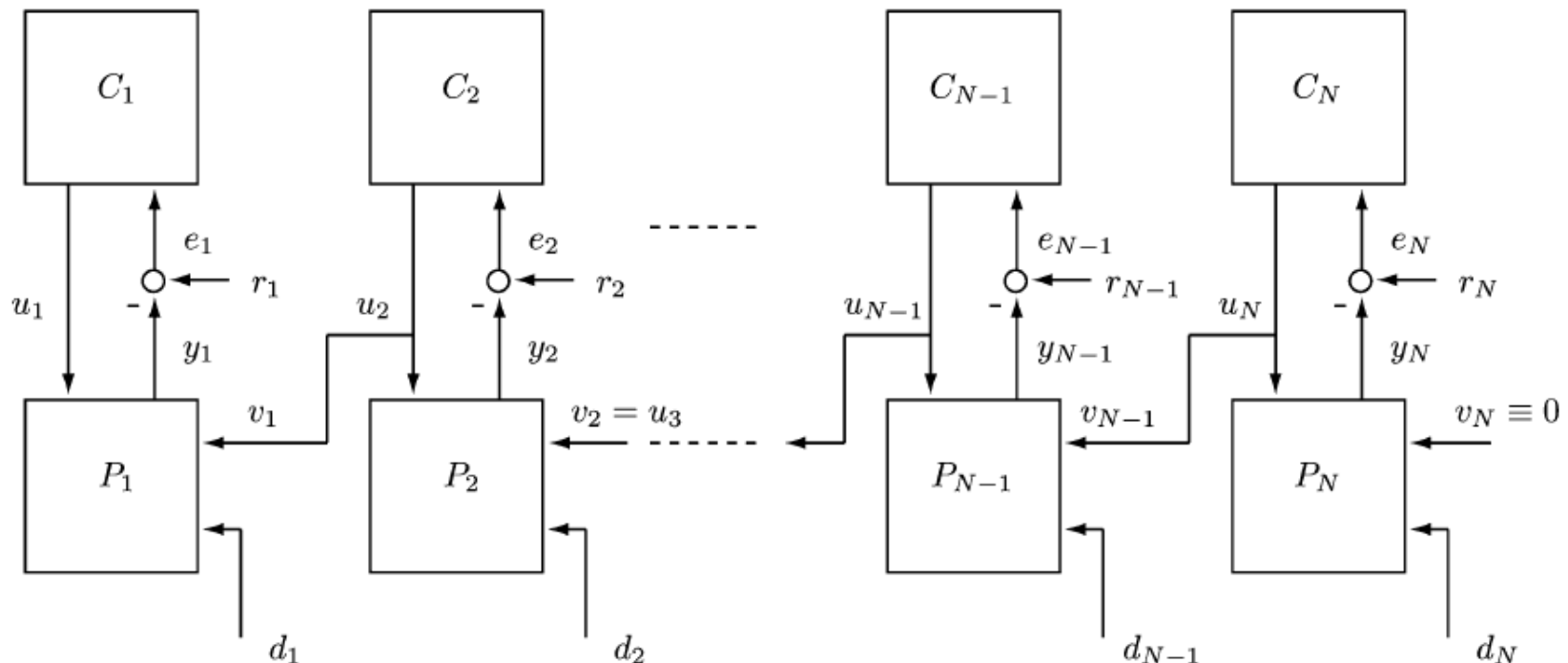


# Examples

## Irrigation networks

- Local computation
- Decentralized information structure

Actions, taken at local pool levels, are computed locally based on local information

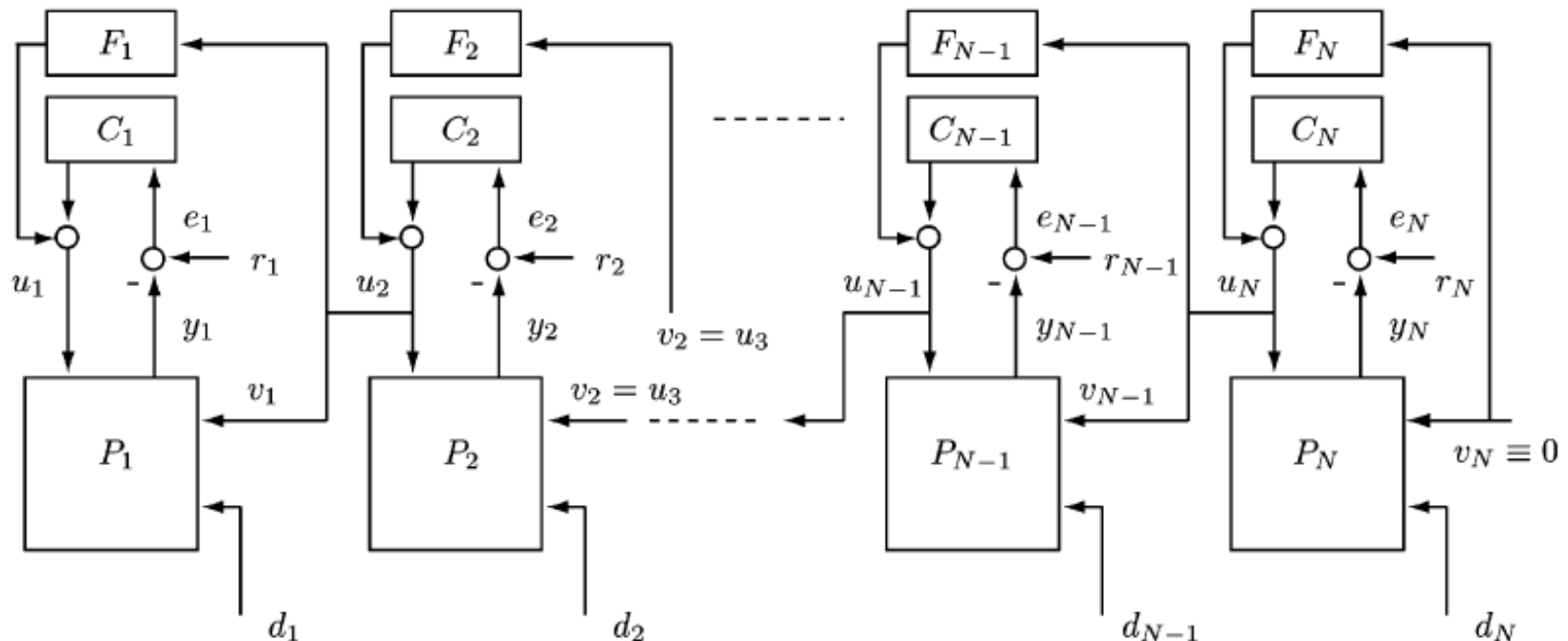


# Examples

## Irrigation networks

- Local computation
- Distributed information structure

Actions, taken at local pool levels, are computed locally based on local information + information transmitted by neighboring gates' control systems

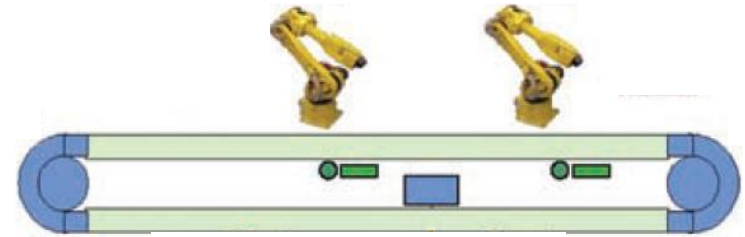


# Examples

## Manufacturing plants

### Serial-parallel manufacturing line:

machining cells connected by conveyors



### Manufacturing plants:

Large quantities of data: **thousands of sensors** measure positions, velocities, material flows, temperatures, **hundreds of times every minute.**

### Main functions:

- **Control:**
  - **Device-level control** (servo-systems, flow, and temperature). ⇒ Real-time communication requirements (small but frequently-transmitted packets) – CAN BUS
  - **Subsystem-level control**, to enforce cooperation between more devices. ⇒ Coordination signal exchange - SCADA
  - **Factory-level control**: coordinates multiple cells. Links the factory floor control infrastructure to the enterprise-level systems (part ordering, supply-chain integration). ⇒ No real-time requirements – Ethernet

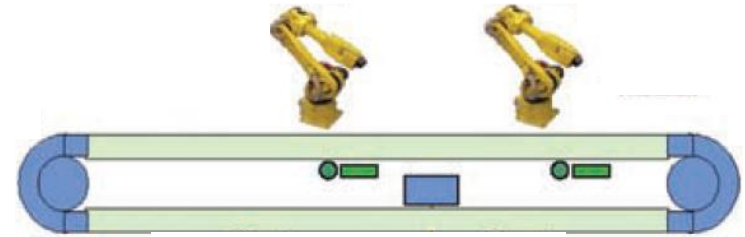
From: J. R. Moyne and D. M. Tilbury. The emergence of industrial control networks for manufacturing control, diagnostics, and safety data. *Proceedings of the IEEE*, 95 (1), pp. 29-47

# Examples

## Manufacturing plants

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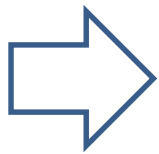


### Manufacturing plant main functions:

- **Diagnostics:**
  - Schedule a tool change
  - Shut down an underperforming tool

⇒ Large amount of data sent infrequently - Ethernet
- **Safety:** safety systems and safety functions must be reconfigurable and coordinated between each other 

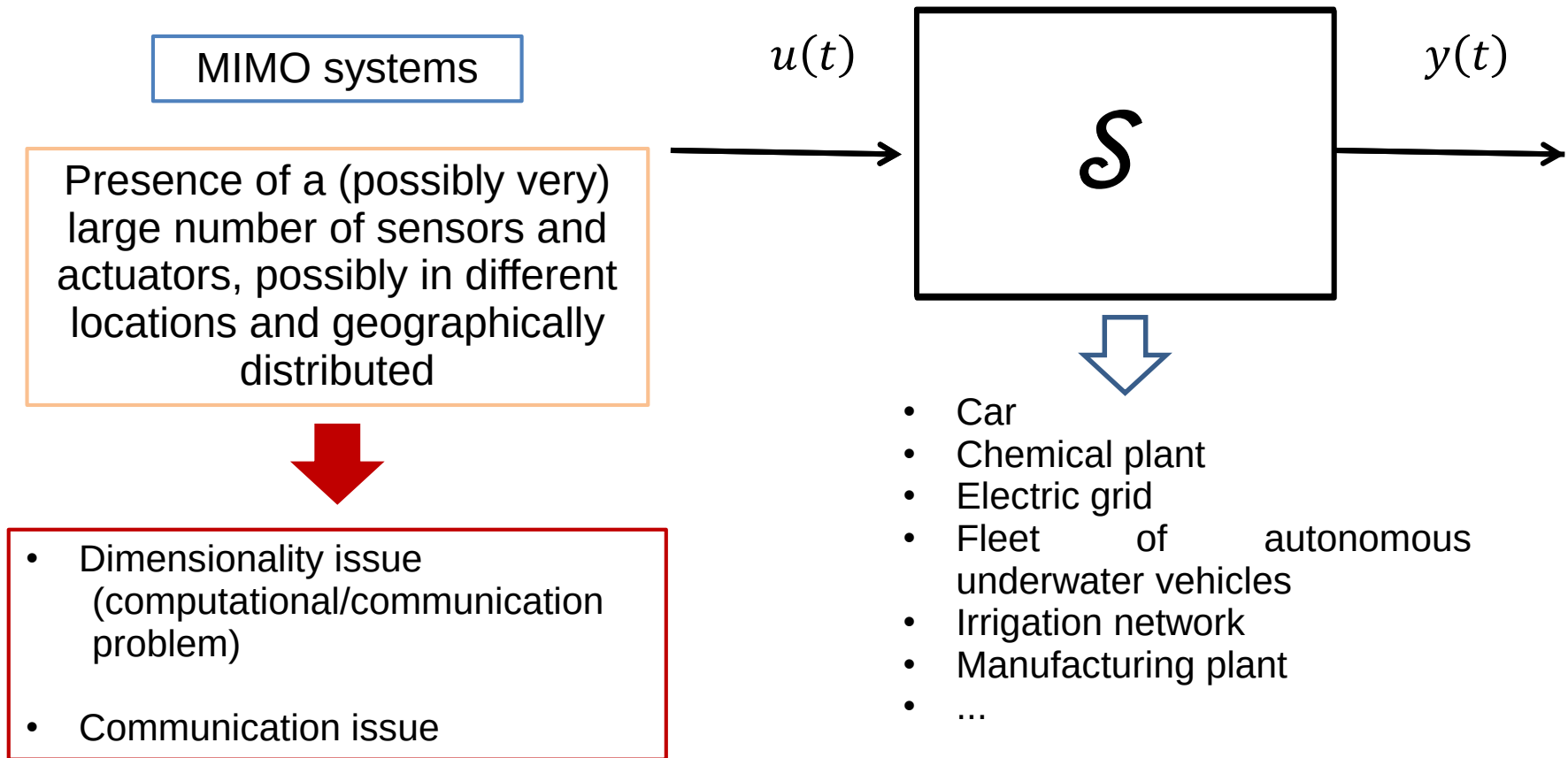
⇒ Safety functions must be **guaranteed** within a **defined time**. High level of determinism



A strongly **hierarchical control architecture** must be employed with different communication networks at different layers to guarantee the fulfillment of the different functions with different timings and priorities.

# Networked control

**What are the common features of the examples?**



# Networked control

## What are the common features of the examples?

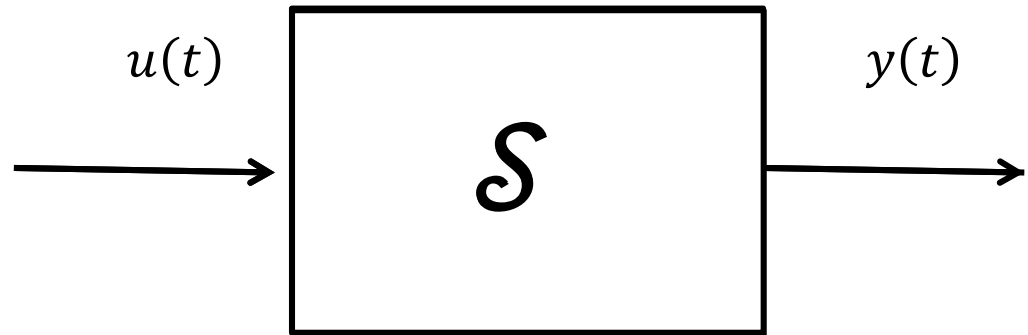
### MIMO systems

Very complex and large-scale system under control

- Distributed/multilayered system structure
- Large number of interacting subsystems



- High-dimensional and complex control problem.
- Need of modular and hierarchical control scheme
  - Subsystem/device level
  - Coordination between subsystems (which are often regarded as actuators for higher-level control systems).



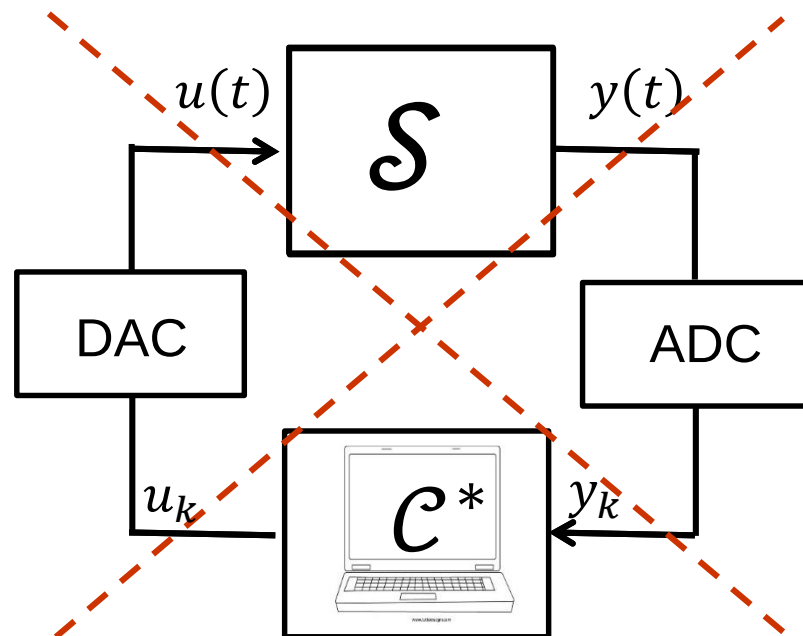
- Car
- Chemical plant
- Electric grid
- Fleet of autonomous underwater vehicles
- Irrigation network
- Manufacturing plant
- ...

# Networked control

## What are the common features of the examples?

New control applications pose new **challenges**.

Classic «centralized» control solutions with ideal communication may be not feasible/possible/adequate





# Networked control

## Technological advances:

**Low-cost processing power.**

Possibility of controlling at remote locations and/or at subsystem level.

Possibility of transmitting information reliably via shared digital networks (even wirelessly).



## Emergence of new areas of research:

**Complex control architectures (distributed, hierarchical) decentralized,**

**Control over dedicated (serial) communication networks**

# Networked control

This course will deal with these two topics:

- **Complex control architectures** (distributed and decentralized)
- **Control over** dedicated (serial) communication **networks**

and in particular with their

| opportunities                                                                                                                                                                             | challenges                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- Flexibility</li><li>- Reliability</li><li>- Robustness</li><li>- Reconfigurability</li><li>- Scalability (computational, communication)</li></ul> | <ul style="list-style-type: none"><li>- Control challenges (design of distributed, decentralized, hierarchical controllers)</li><li>- Network-induced challenges (bandwidth limitations, delays, jitter, packet dropouts)</li></ul> |

## Main related references

- G.P. Ferrari, E. Pizzi, Applicazione di schemi innovativi di controllo predittivo alla sezione di refrigerazione di un impianto per il trattamento del gas naturale, *MSc Thesis*, Politecnico di Milano (supervisor: Prof. M. Farina), 2013-2014.
- M. Banfi, Sviluppo e analisi di sistemi di controllo per l'inseguimento di segnali applicati a un impianto chimico di trattamento di gas naturale, *MSc Thesis*, Politecnico di Milano (supervisor: Prof. M. Farina), 2014-2015.
- R. J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28.
- J. R. Moyne and D. M. Tilbury. The emergence of industrial control networks for manufacturing control, diagnostics, and safety data. *Proceedings of the IEEE*, 95 (1), pp. 29-47.
- N. E. Leonard *et al.* Collective motion, sensor networks, and ocean sampling. *Proceedings of the IEEE*, 95 (1), pp. 48-74.
- M. Cantoni *et al.* Control of large-scale irrigation networks. *Proceedings of the IEEE*, 95 (1), pp. 75-91.